(FP2) 5. Complex numbers

NO. 1

a. Use de Moivre's theorem to show that $\cot 6\theta = \dfrac{\cot^6\theta - 15\cot^4\theta + 15\cot^2\theta - 1}{6\cot^5\theta - 20\cot^3\theta + 6\cot\theta}$. [6]

b. Hence obtain the roots of the equation

$$x^6 - 6x^5 - 15x^4 + 20x^3 + 15x^2 - 6x - 1 = 0$$

in the form $\cot(q\pi)$, where q is a rational number. [4]

A level FURTHER MATHEMATICS 2024/Winter/Paper2_2/Q4

Mark scheme

Question	Answer	Marks	Guidance
a	$\cos 6\theta = \text{Re}(\cos\theta + i\sin\theta)^6 = \cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta$	M1A1	Expands and takes real part.
	$\cos 6\theta = \text{Im}(\cos\theta + i\sin\theta)^6 = 6\cos^5\theta\sin\theta - 20\cos^3\theta\sin^3\theta + 6\cos\theta\sin^5\theta$	M1A1	Takes imaginary part
	$\cot 6\theta = \dfrac{\cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta}{6\cos^5\theta\sin\theta - 20\cos^3\theta\sin^3\theta + 6\cos\theta\sin^5\theta} \times \dfrac{\sin^{-6}\theta}{\sin^{-6}\theta}$	M1	Divides numerator and denominator by $\sin^6\theta$.
	$\cot 6\theta = \dfrac{\cot^6\theta - 15\cot^4\theta + 15\cot^2\theta - 1}{6\cot^5\theta - 20\cot^3\theta + 6\cot\theta}$	A1	AG
		6	
b	$x = \cot\theta$, $\cot 6\theta = 1$	B1	Applies identity given in **(b)**.
	$6\theta = \frac{1}{4}\pi + k\pi$	M1	Solves $\cot 6\theta = 1$
	$\cot\left(\frac{1}{24}\pi\right)$	A1	Gives one correct solution. 7.60
	$\cot\left(\frac{5}{24}\pi\right), \cot\left(\frac{3}{8}\pi\right), \cot\left(\frac{13}{24}\pi\right), \cot\left(\frac{17}{24}\pi\right), \cot\left(\frac{7}{8}\pi\right)$	A1	Gives other five solutions. 1.30, 0.41, –0.13, –0.77, –2.41 Withhold A1 mark for repeated roots.
		4	

Comment

a. After expanding $(\cos\theta + i\sin\theta)^6$ isin using the binomial expansion, most candidates accurately grouped together terms contributing to the real and imaginary parts. Good responses then showed clear steps when dividing by $\sin^6\theta$ to fully justify the given result.

b. Almost all candidates rearranged the polynomial equation correctly, linking with the previous part of the question by substituting $x = \cot\theta$. When solving $\cot 6\theta = 1$, most candidates obtained $x = \cot\left(\frac{1}{24}\pi\right)$. A few candidates gave the values for θ rather than actual solutions as instructed in the question.

NO. 2

Find the roots of the equation $z^3 = -108\sqrt{3} + 108\mathrm{i}$, giving your answers in the form $r(\cos\theta+\mathrm{i}\sin\theta)$, where $r > 0$ and $0 < \theta < 2\pi$. [5]

A level FURTHER MATHEMATICS 2024/Summer/Paper2_1/Q1

Mark scheme

Question	Answer	Marks	Guidance
1	$z^3 = 216\cos\frac{5\pi}{6} + \mathrm{i}216\sin\frac{5\pi}{6}$	B1	Finds modulus and argument of $-108\sqrt{3} + 108\mathrm{i}$
	$z_1 = 6\cos\frac{5\pi}{18} + \mathrm{i}6\sin\frac{5\pi}{18}$	M1 A1FT	Finds one root. FT on their modulus. For M1 must have divided their argument of $-108\sqrt{3} + 108\mathrm{i}$ by 3.
	$z_2 = 6\cos\frac{17\pi}{18} + \mathrm{i}6\sin\frac{17\pi}{18}$, $z_3 = 6\cos\frac{29\pi}{18} + \mathrm{i}6\sin\frac{29\pi}{18}$	M1 A1	Finds other two roots.
		5	

Comment

Good candidates showed clear working, starting from z^3 with the correct argument and listing all three roots in the required form. A significant number of candidates used $-\frac{1}{6}\pi$ as the argument of z^3 instead of $\frac{5}{6}\pi$. This led to incorrect answers when they attempted to take the cube root. A simple diagram would have aided in visualising that the argument is in the second quadrant, not the fourth.

NO. 3

a. Use de Moivre's theorem to show that

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta.$$ [4]

b. Hence obtain the roots of the equation

$$32x^5 - 40x^3 + 10x - \sqrt{2} = 0$$

in the form $\cos(q\pi)$, where q is a rational number. [4]

A level FURTHER MATHEMATICS 2023/Winter/Paper2_2/Q3

Mark scheme

Question	Answer	Marks	Guidance
a	$\cos5\theta = \text{Re}(\cos\theta + i\sin\theta)^5 = \cos^5\theta - 10\sin^2\theta\cos^3\theta + 5\sin^4\theta\cos\theta$	M1 A1	Expands and takes real part. Accept RHS to LHS using $2\cos\theta = z + \frac{1}{z}$.
	$= \cos^5\theta - 10\cos^3\theta(1 - \cos^2\theta) + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta)$	M1	Applies $\cos^2\theta = 1 - \cos^2\theta$.
	$= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$	A1	AG
		4	
b	$x = \cos\theta, \quad \cos5\theta = \frac{1}{2}\sqrt{2}$	M1	Applies identify given in **(b)**.
	$5\theta = \pm\frac{1}{4}\pi + 2k\pi$	M1	Solves $\cos5\theta = \frac{1}{2}\sqrt{2}$
	$\cos\left(\frac{1}{20}\pi\right)$	A1	Gives one correct solution. Accept $q = \frac{1}{20}$.
	$\cos\left(\frac{9}{20}\pi\right), \cos\left(\frac{17}{20}\pi\right), \cos\left(\frac{25}{20}\pi\right), \cos\left(\frac{33}{20}\pi\right)$	A1	Gives other solutions. OE. A0 for repeated roots.
		4	

Comment

a. After expanding $(\cos\theta + i\sin\theta)^5$ using the binomial expansion, most candidates grouped together terms contributing to the real part before applying the identity $\cos^2\theta + \sin^2\theta = 1$ to fully justify the given result. Alternatively, a few candidates worked from the right-hand side to the left-hand side, applying $2\cos\theta = z + z^{-1}$, and were usually successful too.

b. Almost all rearranged the polynomial equation correctly, linking with the previous part of the question by substituting $x = \cos\theta$. Solving $\cos5\theta = \frac{1}{2}\sqrt{2}$, most candidates obtained $x = \cos\left(\frac{1}{20}\pi\right)$. A few candidates gave the values for θ rather than actual

solutions, highlighting the need to read the question carefully. Although the four other solutions were often given in different ways, few candidates gave repeated solutions.

NO. 4

Find the roots of the equation $(z+5\mathrm{i})^3 = 4+4\sqrt{3}\,\mathrm{i}$, giving your answers in the form $r\cos\theta + \mathrm{i}(r\cos\theta - 5)$, where $r > 0$ and $0 < \theta < 2\pi$. [5]

A level FURTHER MATHEMATICS 2023/Winter/Paper2_1/Q2

Mark scheme

Question	Answer	Marks	Guidance
2	$(z+5\mathrm{i})^3 = 4+4\mathrm{i}\sqrt{3} = 8\mathrm{e}^{\frac{1}{3}\pi}$	B1	Finds modulus and argument of $4+4\mathrm{i}\sqrt{3}$
	$z_1 = 2\left(\cos\frac{1}{9}\pi + \mathrm{i}\sin\frac{1}{9}\pi\right) - 5\mathrm{i} = 2\cos\frac{1}{9}\pi + \mathrm{i}\left(2\sin\frac{1}{9}\pi - 5\right)$	M1 A1	Finds one root.
	$z_2 = 2\cos\frac{7}{9}\pi + \mathrm{i}\left(2\sin\frac{7}{9}\pi - 5\right)$, $z_3 = 2\cos\frac{13}{9}\pi + \mathrm{i}\left(2\sin\frac{13}{9}\pi - 5\right)$	A1 FT	Finds other two roots. FT on their modulus.
		A1 FT	
		5	

Comment

The majority of candidates correctly stated the modulus and argument of $4+4\sqrt{3}\mathrm{i}$. Good candidates maintained accuracy when taking the cube root of the modulus and manipulating the argument, leaving the three roots in the required form.

NO. 5

a. By considering the binomial expansion of $(z+z^{-1})^4$, where $z = \cos\theta + i\sin\theta$, use de Moivre's theorem to show that $\cos^4\theta = \frac{1}{8}(\cos 4\theta + 4\cos 2\theta + 3)$. [5]

b. Use the substitution $x = \sin\theta$ to find the exact value of $\int_0^{\frac{1}{2}} (1-x^2)^{\frac{3}{2}} \mathrm{d}x$. [3]

A level FURTHER MATHEMATICS 2023/Summer/Paper2_1/Q3

Mark scheme

Question	Answer	Marks	Guidance
a	$z+z^{-1} = 2\cos\theta$	B1	Use of $z+z^{-1} = 2\cos\theta$.
	$(z+z^{-1})^4 = (z^2 + z^{-4}) + 4(z^2 + z^{-2}) + 6$	M1 A1	Expands and groups. M1 A0 for no clear grouping. Correct substitution of $z^n = \cos n\theta + i \sin n\theta$ for each term scores M1 A1.
	$(2\cos\theta)^4 = 2\cos 4\theta + 4(2\cos 2\theta) + 6$	M1	Substitutes $z^n+z^{-n} = 2\cos n\theta$. If $z^n = \cos n\theta + i \sin n\theta$ used then must cancel sin.
	$\cos^4\theta = \frac{1}{8}(\cos 4\theta + 4\cos 2\theta + 3)$	A1	AG. SC B1 expands $(\cos\theta + i \sin\theta)^4$ and uses trigonometric identities.
		5	
b	$\int_0^{\frac{1}{2}} \left(1-x^2\right)^{\frac{3}{2}} \mathrm{d}x = \int_0^{\frac{1}{6}\pi} \cos^4\theta \,\mathrm{d}\theta = \frac{1}{8}\left[\frac{1}{4}\sin 4\theta + 2\sin 2\theta + 3\theta\right]_0^{\frac{1}{6}\pi}$	M1 A1	Applies substitution (M1) gets to $\int \cos^4\theta \, d\theta$, changes limits, integration correct (A1).
	$\frac{1}{16}\left(\frac{9}{4}\sqrt{3} + \pi\right)$	A1	
		3	

Comment

a. After expanding $(z + z^{-1})^4$ using the binomial expansion, good candidates grouped together terms clearly before applying the identity $z^n + z^{-n} = 2\cos n\theta$ to fully justify their answer. For the left-hand side, the majority of candidates clearly applied $z + z^{-1} = 2\cos\theta$.

b. Most candidates successfully applied the given substitution to change the integral to $\int \cos^4 \theta d\theta$. Good candidates maintained accuracy when changing limits and integrating using **part (a)**.

NO. 6

a. Write down the fourth roots of unity. [1]

b. Use de Moivre's theorem to show that $\cos 4\theta = 8\cos 4\theta - 8\cos 2\theta + 1$. [4]

c. Hence obtain the real roots of the equation

$$16(8x^4 - 8x^2 + 1)^4 - 9 = 0$$

in the form $\cos(q\pi)$, where q is a rational number. [5]

A level FURTHER MATHEMATICS 2022/Winter/Paper2_2/Q5

Mark scheme

Question	Answer	Marks	Guidance
a	±1, ±i	B1	Accept exponential form $e^{0}, e^{i\frac{\pi}{2}}, e^{i\pi}, e^{i\frac{3\pi}{2}}$ or trigonometric form. Four values specified.
		1	
b	$\cos 4\theta = \text{Re}(\cos\theta + i\sin\theta)^4 = \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta$	M1 A1	Expands and takes real part.
	$\cos^4\theta - 6\cos^2\theta(1-\cos^2\theta)+(1-2\cos^2\theta + \cos^4\theta)$	M1	Applies $\sin^2\theta = 1 - \cos^2\theta$.
	$8\cos^4\theta - 8\cos^2\theta + 1$	A1	AG. Can award final A1 without previous A1 if penultimate line seen. SC. Applying $\cos 2A = 2\cos^2 A - 1$ twice scores 1/4.
		4	
c	$\left(8x^4 - 8x^2 + 1\right)^4 = \frac{9}{16}, \quad x = \cos\theta$	M1	Applies identify given in (b).
	$\cos 4\theta = \pm\frac{1}{2}\sqrt{3}$	A1	Need ±and exact value for A1.
	$4\theta = \pm\frac{1}{6}\pi + 2k\pi \quad 4\theta = \pm\frac{5}{6}\pi + 2k\pi$	M1	Solves $\cos 4\theta = \frac{1}{2}\sqrt{3}$ or $\cos 4\theta = -\frac{1}{2}\sqrt{3}$
	$\cos\left(\frac{1}{24}\pi\right)$	A1	Gives one correct solution.
	$\cos(\frac{11}{24}\pi), \cos(\frac{13}{24}\pi), \cos(\frac{23}{24}\pi),$ $\cos(\frac{5}{24}\pi), \cos(\frac{7}{24}\pi), \cos(\frac{17}{24}\pi), \cos(\frac{19}{24}\pi),$	A1	Gives other solutions. Accept $\pm\cos\left(\frac{1}{24}\pi\right), \pm\cos(\frac{11}{24}\pi), \pm\cos(\frac{5}{24}\pi), \pm\cos(\frac{7}{24}\pi)$
		5	

Comment

a. Almost all candidates wrote down the four fourth roots of unity, often in exponential or trigonometric form.

b. After expanding $(\cos\theta + i\sin\theta)^4$ using the binomial expansion, most candidates grouped together terms contributing to the real part before applying the identity $\cos^2\theta + \sin^2\theta = 1$ to fully justify the given result. Alternatively, a few candidates worked from the right-hand side to the left-hand side, applying $2\cos\theta = z + z^{-1}$, and were usually successful too.

c. Almost all candidates factorised correctly, linking with the previous part of the question. Stronger candidates then solved $\cos 4\theta = \pm\frac{1}{2}\sqrt{3}$, remembering to consider both signs, leading to exactly eight distinct real roots.

NO. 7

a. Use de Moivre's theorem to show that

$$\operatorname{cosec} 7\theta = \frac{\operatorname{cosec}^7\theta}{7\operatorname{cosec}^6\theta - 56\operatorname{cosec}^4\theta + 112\operatorname{cosec}^2\theta - 64}.$$ [6]

b. Hence obtain the roots of the equation

$$x^7 - 14x^6 + 112x^4 - 224x^2 + 128 = 0$$

in the form cosec $q\pi$, where q is rational. [5]

A level FURTHER MATHEMATICS 2022/Summer/Paper2_1/Q7

Mark scheme

Question	Answer	Marks	Guidance
a	$\sin 7\theta = \mathrm{Im}(c+is)^7$	M1	Uses binomial theorem.
	$\sin 7\theta = -s^7 + 21c^2s^5 - 35c^4s^3 + 7c^6s$	A1	
	$\sin 7\theta = -s^7 + 21(1-s^2)s^5 - 35(1-s^2)^2s^5 + 7(1-s^2)^3s$ $\sin 7\theta = -s^7 + 21(1-s^2)s^5 - 35(1-2s^2+s^4)s^3 + 7(1-3s^2+3s^4-s^6)s$	M1	Uses $c^2 = 1 - s^2$.
	$\sin 7\theta = 64s^7 + 112s^5 - 56s^3 + 7s$	A1	
	$\mathrm{cosec}\, 7\theta = \dfrac{1}{-64s^7 + 112s^5 - 56s^3 + 7s}$	M1	Divides numerator and denominator by s^7.
	$\mathrm{cosec}\, 7\theta = \dfrac{\mathrm{cosec}^7\theta}{7\mathrm{cosec}^6\theta - 56\mathrm{cosec}^4\theta + 112\mathrm{cosec}^2\theta - 64}$	A1	AG CWO
		6	
b	$x^7 = 2(7x^6 - 56x^4 + 112x^2 - 64)$ leading to $\dfrac{x^7}{7x^6 - 56x^4 + 112x^2 - 64} = 2$	M1 A1	Relates with equation in part (a).
	$\mathrm{cosec}7\theta = 2$ leadi to $\sin 7\theta = \frac{1}{2}$	M1	Solves $\sin 7\theta = \frac{1}{2}$
	$x = \mathrm{cosec}\left(\frac{1}{42}\pi\right)$	A1	Gives one correct solution.
	$x = \mathrm{cosec}(q\pi)$, $q = -\frac{19}{42}, -\frac{11}{42}, -\frac{7}{42}, \frac{5}{42}, \frac{13}{42}, \frac{17}{42}$	A1	Gives six other solutions. Allow different values of q as long as all seven solutions are found.
		5	

Comment

a. Most candidates used de Moivre's theorem successfully to find the required $\text{cosec}7\theta$. A few candidates had difficulty in translating their expression in cos and sin into an expression in cosec. There were some elegant uses of $\cot^2\theta = \text{cosec}^2\theta - 1$ and the best responses showed full details of all the steps in the working. This was particularly important since the answer was given in the question.

b. Better responses made the connection with the equation given in **part (a)**, justifying the relationship, then used $\text{cosec}7^2\theta =$ to find all seven distinct solutions.